

Responses to Reviewers

Paper ID: 3205

Action Editor: Yansong Feng

Revision date: 19 September 2026

Title: HIDE and Seek: Detecting Hallucinations in Language Models via Decoupled Representations

Dear Action Editor and Reviewer,

Thank you for your careful review and helpful suggestions. We address each comment below. Changes in the paper are marked in magenta using `\revisionr3`. The locations below refer to the revised paper.

Summary of changes

- Added attention mass and an input-derived norm as standalone detection baselines, evaluated alongside HIDE on the same answers.
- Updated Figure 2 and Section 4.4 for both Llama-3-8B and Gemma-2-9B, and qualified the interpretation of their correlations.
- Corrected the inconsistent latency example and made the timing figures, tables, and percentages agree.
- Added Gemma-2-27B to the timing analysis and limited the scaling claim to the tested models and the stated score complexity.
- Corrected the listed typos, made the requested captions self-contained, and explained the limits of the new results.

Responses to the Action Editor

AE1. Consistency of latency numbers

There are inconsistent descriptions regarding latency time in Section 6.4 and Figure 4.

Response. We found that the example mixed SQuAD and RACE timings. For Llama-3-8B on SQuAD, the timing sheet gives 1.6169 s for base generation and 1.8445 s for HIDE, an overhead of 0.2276 s (14.08%). The 1.7107 s HIDE value belongs to RACE, whose base is 1.5021 s. We corrected the example and regenerated Figure 3 from the sheet.

The separate paired timing experiment on A100 PCIe gives 0.9277 s base and 1.1587 s HIDE total on Llama-3-8B/SQuAD, an overhead of 0.2310 s (24.90%). Table 6 and Figure 4 use these same paired measurements. The text now identifies the hardware and protocol of each experiment, and never combines their values. Differences and percentages are calculated before rounding.

Location in the revised paper: Section 6.4; Figures 3 and 4; Table 6.

AE2. Section 4.4 and Gemma-2-9B

Rephrase the claims in Section 4.4 and explicitly discuss Gemma-2-9B in the main text.

Response. We revised Section 4.4 to describe what the correlations show without treating them as a causal explanation. Figure 2 now uses the same answers as the new detector comparison. Gemma-2-9B has positive HIDE–norm correlations on SQuAD and NQ: +0.126 and +0.215. Llama-3-8B has +0.003 and +0.367, respectively. These values replace the previous figure’s results, which used different stopping and example-selection conditions. We explain this protocol change in Section 5. The results do not support a general inverse relationship between update magnitude and precision. We also clarify that the measured norm omits value and output projections and is a proxy for the input-derived update.

Location in the revised paper: Section 4.4 and Figure 2; Section 5, “Direct comparison with attention and norm baselines.”

AE3. Model-size scaling

Add at least one substantially larger model or soften the model-size scalability claim.

Response. We added Gemma-2-27B. Its mean added overhead is 0.1969 s on SQuAD (95% confidence interval [0.1644, 0.2365]) and 0.0806 s on NQ ([0.0751, 0.0864]). We also narrowed the claim: the measurements show that HIDE can run at these tested scales, but do not establish constant overhead across models. The tested hidden widths span 3,072–4,608, and answer lengths differ. The $\mathcal{O}(d)$ statement is retained only for computing the score at a fixed number of selected tokens, not for total pipeline latency.

Location in the revised paper: Section 6.4, “Model scale and complexity”; Figure 4; Tables 6 and F.4.

AE4. Standalone detection baselines

Add attention mass and input-derived update norm as standalone detection baselines, with discussion.

Response. We added AUC and PCC for HIDE, attention mass, and the input-derived norm on the same 2,000 examples per dataset and model, covering Llama-3-8B and Gemma-2-9B on all four QA datasets. Attention has higher AUCs in seven of eight settings, including all four closed-book factuality settings. HIDE performs better on Gemma-2-9B/RACE. We report both outcomes and discuss their implications.

Panel A of the new tables repeats the full-benchmark HIDE values from Appendix B for reference. Panel B compares all three detectors on identical answers under the first-answer-line stopping conditions. The captions explain why values from the two panels cannot be compared directly.

Location in the revised paper: Section 5; Section 6.2 and Table 4; Appendix F, Tables F.1–F.3.

Responses to Reviewer 1

Summary and strengths

Thank you for recognizing the motivation, single-pass design, evaluation, and ablations. We have retained these experiments and revised the interpretation where the additional comparisons call for it.

Weakness 1. Value beyond attention mass

Would a cheaper attention-mass detector match HIDE? Report attention mass and the input-derived norm as standalone baselines.

Response. The direct comparison shows that attention mass is a strong baseline. HIDE performs better on Gemma-2-9B/RACE, with an AUC_s advantage of 6.69 percentage points (95% confidence interval [3.70, 9.69]); the advantage also holds with ROUGE-L. Attention performs better in the other seven settings. We therefore present HIDE as one useful detection option, without claiming that HSIC is necessary in every setting.

HIDE uses hidden states without requiring attention weights. This can matter when an implementation exposes hidden states but not attention weights. We discuss this distinction using Lookback Lens, the analysis of attention and value-vector norms by Kobayashi et al., and FlashAttention. Both HIDE and our raw attention baseline are training-free. We do not claim a measured speed advantage over attention or an untested benefit from combining the detectors.

We also report a selected-token-count control, which closely tracks HIDE. This limits attributing the score’s performance specifically to nonlinear dependence, and is discussed in the paper.

Location in the revised paper: Section 6.2, “Matched comparison with attention and norm baselines” and “Interpretation of the baseline comparison”; Table 4; Tables F.1–F.3; Limitations.

Weakness 2. Model-dependent magnitude versus precision

The negative HIDE–norm correlation for Llama reverses sign for Gemma, so the general interpretation is not supported.

Response. We agree and have removed the general inverse interpretation. The main text now reports both models explicitly. Gemma’s norm correlations are positive, and the earlier negative Llama/SQuAD correlation is not reproduced under the first-answer-line protocol. We state that the relationship depends on the model, dataset, and evaluation conditions. The QCBW comparison remains a way to examine incorrect answers that are close to the question in topic; it is not treated as causal evidence.

Location in the revised paper: Section 4.4 and Figure 2; Section 5 for the stopping conditions.

Weakness 3. Efficiency and limited scaling evidence

The efficiency numbers are internally inconsistent, and two nearby hidden dimensions provide limited evidence for substantially larger models.

Response. We corrected the dataset mix-up described in AE1 and identified the two timing experiments clearly. Figure 3 reports the baseline comparison; Figure 4 reports paired overhead measurements for four models, including Gemma-2-27B. We removed the constant-overhead claim and distinguish score complexity from total runtime.

Location in the revised paper: Section 6.4; Figures 3 and 4; Tables 6 and F.4.

Substantive encouragement. AUC/PCC, especially for factuality

Report attention and norm alongside HIDE, especially on NQ and TriviaQA; discuss settings where HIDE does not retain an advantage.

Response. We report AUC_s , PCC_s , AUC_r , and PCC_r for all three detectors on the same answers. Attention performs better on both NQ and TriviaQA for both models. For example, TriviaQA AUC_s is 57.01 for HIDE versus 73.68 for attention on Llama, and 49.83 versus 66.94 on Gemma. The norm also performs better than HIDE on both TriviaQA settings. The discussion explicitly recognizes attention to the question as a useful signal even without a supplied evidence passage.

For this comparison, generation stops after the first nonempty answer line, avoiding continuations into another question. All three detectors were computed on these same generations. We retain all 16,000 selected records, including capped and empty answers and cases without output states. The sample size was chosen for the available compute time after earlier runs had been inspected; examples were not selected by detector performance. The full-benchmark results remain separate because their stopping conditions differ.

The additional control uses $(n - 1)/n^2$, where n is the actual selected-token count, and zero when no output states are available. Its PCC with HIDE is at least 0.9987, and its AUC_s differs by less than 0.43 percentage points in every setting. We report this limitation on interpreting the score as evidence of nonlinear dependence. We also clarify the sampling assumptions of the asymptotic result in Appendix A.3 and make Algorithm 1 and Section 4.3.1 explicit about cached states, keyword matching, and the zero-score fallback, in agreement with the implementation.

Location in the revised paper: Algorithm 1 and Section 4.3.1; Section 5; Section 6.2 and Table 4; Appendix A.3; Appendix F, Tables F.1–F.3; Limitations.

Required minor revision 1. Reconcile efficiency values

Correct the inconsistent complete-pipeline and overhead values and ensure Section 6.4, Figure 4, and the response letter agree.

Response. Corrected as detailed in AE1. The benchmark Llama-3-8B/SQuAD example is now 1.6169 s base, 1.8445 s total, and 0.2276 s overhead (14.08%). The separate paired experiment supplies both Table 6 and Figure 4. It uses 200 queries per model/dataset, ten warm-up pairs, and three measured base/HIDE pairs per query. Repeats are averaged within each query before

reporting means and confidence intervals. Total latency includes generation with hidden-state capture, keyword extraction, scoring, and bookkeeping; loading and correctness evaluation are excluded. All measurements are retained. These timings use the benchmark stopping conditions.

Location in the revised paper: Section 6.4; Table 6; Figure 4.

Required minor revision 2. Explicit Gemma correlations

Qualify the magnitude-versus-precision interpretation and explicitly state Gemma-2-9B’s positive update-norm correlations.

Response. Section 4.4 and Figure 2 now give Gemma’s correlations as +0.126 on SQuAD and +0.215 on NQ, and Llama’s as +0.003 and +0.367, respectively. The text explains that a larger norm does not by itself establish more precise use of input information. We removed the explanation that attributed hallucination to noisy updates based on a negative correlation.

Location in the revised paper: Section 4.4 and Figure 2.

Encouraged minor revision 1. Consistent efficiency framing

Frame the 51% figure as a standard-HuggingFace baseline comparison; optimized-serving and lower-bound statements must remain hypothetical.

Response. We verified the percentage against the timing sheet. For each model and method, we first average the four dataset latencies equally. We then average the six model-specific percentage reductions. This gives 50.8% relative to EigenScore, 50.8% relative to lexical similarity, and 53.2% relative to LN-Entropy. The paper consistently uses 50.8% relative to five-generation EigenScore under standard HuggingFace inference. We removed guaranteed speedup and lower-bound claims. The discussion of optimized serving is explicitly an implementation argument, with no vLLM or PagedAttention measurements claimed.

Location in the revised paper: Abstract; Section 1 contributions; Section 6.4; Figure 3; Conclusions.

Encouraged minor revision 2. Larger model or narrower claim

Add a substantially larger model or soften scalability claims; retain the scoped $\mathcal{O}(d)$ argument.

Response. We did both: added Gemma-2-27B and narrowed the claim. At fixed selected count n , RBF Gram construction costs $\mathcal{O}(n^2d)$, and the matrix operations add $\mathcal{O}(n^3)$. This is $\mathcal{O}(d)$ in hidden width when n is fixed. It does not describe the cost of generation, keyword extraction, or the complete pipeline. Answer lengths and timing uncertainty are reported so the scope of the comparison is clear.

Location in the revised paper: Section 6.4, “Model scale and complexity”; Figure 4; Tables 6 and F.4.

Typographic point 1

Replace “actuality hallucinations” with “factuality hallucinations” in Section 6.2.

Response. Corrected.

Location in the revised paper: Section 6.2, opening paragraph.

Typographic point 2

Replace “perfromance” with “performance” in the introduction to Section 6.

Response. Corrected and simplified the surrounding sentence.

Location in the revised paper: Section 6, introductory paragraphs.

Typographic point 3

Replace “verbatim repeatation” with “verbatim repetition” in Section 8.2.

Response. Corrected.

Location in the revised paper: Section 8.2, paragraph discussing prompt copying.

Typographic point 4

Verify the apparent empty contribution bullet before “Despite its single-pass nature”.

Response. We checked the rendered list. It has three populated items and no empty bullet. The wording now agrees with the baseline and timing results.

Location in the revised paper: Section 1, contributions list.

Typographic point 5

Confirm that the captions of Figures 3, 5, 6, and 8 are fully self-contained.

Response. We checked and revised all four captions. Figure 3 names the models, datasets, timing platform, averaging rule, and EigenScore comparison. Figures 5, 6, and 8 name the models and datasets, explain what is varied, and define AUC_s using the sentence-similarity correctness threshold. They can be read without relying on the response letter.

Location in the revised paper: Figure 3; Figures 5, 6, and 8 in Sections 7.2–7.5.

We also checked the numerical summaries against the displayed benchmark tables. Without changing their per-dataset values, we corrected the factuality win count to 9/12, its mean relative AUC gain over the better uncertainty baseline to 23.3%, and the faithfulness gain over the best multi-pass baseline to 4.2%. The probe discussion now describes the mixed NQ results. These corrections appear in Sections 6, 6.1, and 6.2.